



Pharm*Achieve*

DIARRHEA

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

BM Bowel Movement

BSS Bismuth Subsalicylate

CDAD *Clostridium Difficile* (*C. difficile*) Associated Diarrhea

CDI *C. difficile* infection

CNS Central Nervous System

ETEC Enterotoxigenic E. Coli

FQ Fluoroquinolones

ID Infectious Diarrhea

ORS/T Oral Rehydration Salts/Therapy

OTC Over-the-Counter

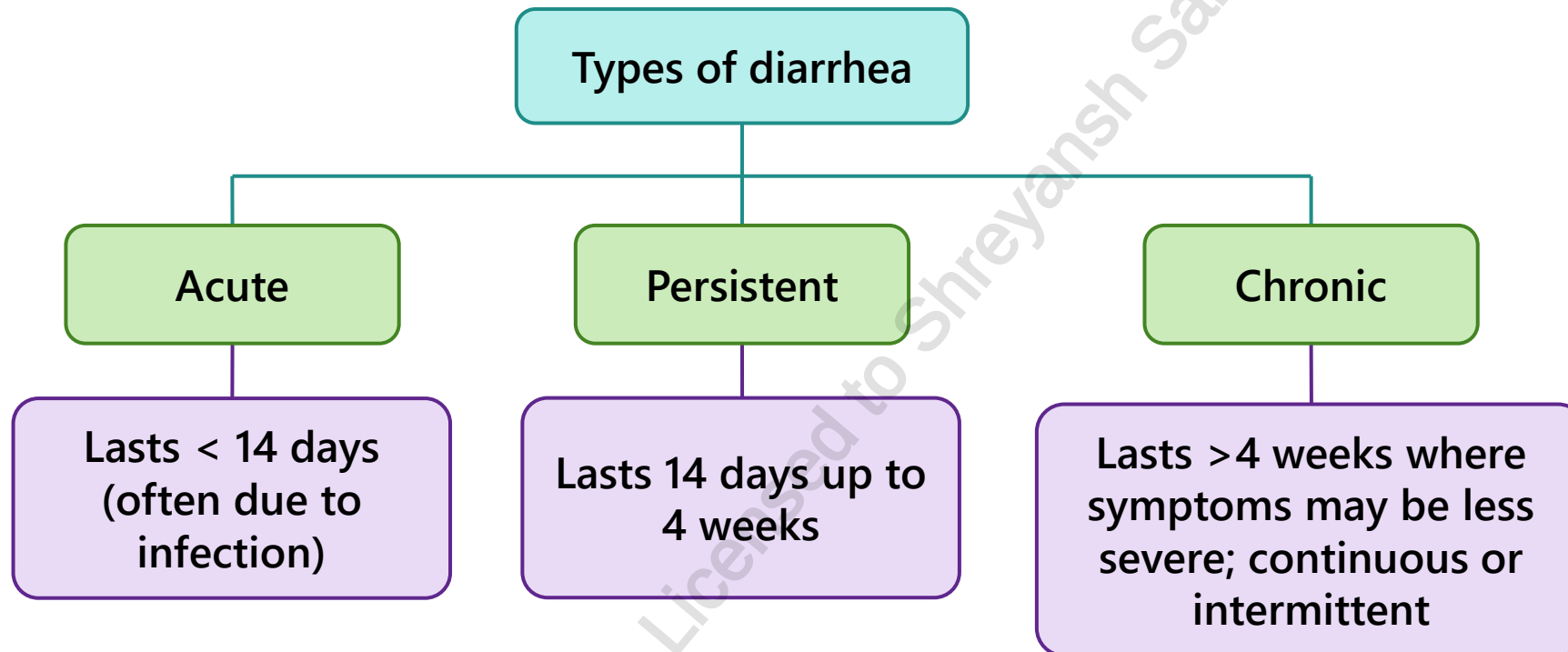
STEC Shiga toxin-producing Escherichia coli

TD Traveler's Diarrhea

WHAT IS DIARRRHEA?

Change in normal bowel movements with the following characteristics:

1. Increased water content
2. Increased volume
3. Increased frequency (≥ 3 loose stools/day)



PATHOPHYSIOLOGY

Diarrhea is a SYMPTOM which may be caused by ANY combination of the following:

Osmotic changes

- Ingestion of a large amount of poorly absorbed solutes pull water into intestinal lumen, resulting in watery stools (Example: lactose intolerance). Fasting improves symptoms.

Secretory changes

- Water secreted into lumen of proximal small intestine and is re-absorbed at distal end
- Diarrhea results when amount of water secreted > than amount reabsorbed (Example: bacteria, some laxatives)

Exudative changes

- Weakened barrier function of intestinal epithelium due to loss of cells (e.g. inflammation)
- Causes leakage of water, blood, electrolytes, mucus, protein or WBCs into lumen

Motility disorders

- Abnormal increase in gut motility
- Abnormal decrease in gut motility (e.g. neuropathy) leading to bacterial overgrowth, which can cause diarrhea

CAUSES

Inflammatory disease causes:

- Crohn's disease
- Ulcerative colitis
- Diverticulitis
- Irritable bowel syndrome
- Ischemic bowel disease
- Ulcerative jejunoileitis

Other causes:

- Malabsorption (e.g. bile acid diarrhea)
- Food poisoning
- Alcohol
- Food intolerances: lactose, sorbitol, fructose, mannitol, xylitol, gluten, excess fiber

DRUG RELATED CAUSES

- **Recent use of antibiotics:** cause benign diarrhea to pseudomembranous colitis (CDI)
 - Clindamycin, cephalosporin (ceftriaxone), carbapenems, quinolones, erythromycin
- **Laxatives:** Lactulose, Senokot®, Na phosphate enema or oral product
- **Magnesium based antacids**
- **Proton-pump inhibitors** (CDI)
- **Histamine-2 receptor antagonists** (CDI)
- **Sorbitol containing drugs**
- **Colchicine**
- **Chemotherapy agents:** 5-fluorouracil, capecitabine, irinotecan → Due to protein losing enteropathy
- **Immunosuppressants:** mycophenolate
- **NSAIDs:** ibuprofen, naproxen → Due to lymphocytic/collagenous colitis
- **Motility agents:** erythromycin, domperidone, metoclopramide
- **Orlistat:** due to malabsorption of fat causing steatorrhea

RISK FACTORS

Age

- < 3 years at risk (daycare attendance increases transmission as well)
- Elderly have high risk of complications due to dehydration

Travel

- Developing countries may have unsafe food/water
- Foods and water may be contaminated with pathogens, causing diarrhea

Others

- Antibiotic use and other medications
- Unsafe sexual practice (risk of shigella infection) may cause diarrhea

RED FLAGS

- Signs and symptoms of dehydration in children and adults
 - Dehydration in children:
 - Dry mouth and/or tongue, little or no tears when crying, < 4 wet diapers in 24 hours, sunken eyes and cheeks, tired and irritable, decreased skin turgor (when pinched and released, the skin flattens slowly)
 - Dehydration in adults:
 - Increased thirst, infrequent urination, darker urine, dry skin, fatigue, dizziness, confusion, light-headedness
- Severe diarrhea (> 6 loose stools per day) for > 48 hours without improvement in adults, > 24 hours without improvement in children
- Blood or abnormal mucous in stool
- Extensive abdominal cramping or pain
- Fever $\geq 38.5^{\circ}\text{C}$
- Persistent or chronic diarrhea
- Persistent vomiting > 4 hours
- Recent use of antibiotics associated with *C. difficile* (e.g. clindamycin, ampicillin, cephalosporins)
- Immunocompromised
- Pregnancy
- Young age (< 2 years old)
- Weight loss due to diarrhea

NON-PHARMACOLOGICAL MEASURES

Oral Rehydration Therapy (ORT)

- ORT for both acute and chronic diarrhea (Gastrolyte® & Pedialyte®)
- Home-made recipe: ½ teaspoonful (2.5 mL) salt + 6 tsp (30 mL) sugar + 4 cups (1 L) of water. If kept at room temperature, discard after 12 hours. If refrigerated, discard after 24 hours
- Contraindicated in patients with protracted vomiting, stupor, coma or intestinal ileus

Avoid products that exacerbate diarrhea

- Alcohol, caffeine, sugary beverages, prune juice
- Avoid lactose if suspected lactose-intolerance
- If infectious, avoid loperamide: prevents removal of toxin from bowel (↑ risk of toxicity)

Report outbreaks

- Help public health recognize pathogens to prevent population spread

Prevent diarrhea

- Administer vaccines (e.g. rotavirus, cholera), frequent hand washing (alcohol-based disinfectants do not kill *C. difficile* spores) and travellers' diarrhea prevention education

NON-PHARMACOLOGICAL MEASURES

Doses are **FYI**

	Rehydration	Maintenance
Mild-moderate dehydration, children	ORS, 50–100 mL/kg over 3–4 hours	<10kg: 60-120 mL ORS per BM (500mL/day) >10kg: 120-240 mL ORS per BM (1L/day)
Mild-moderate dehydration, adolescents and adults	2-4 L	As thirst dictates, up to 2L/day
Severe dehydration, children	10 ml/kg IV crystalloid, repeated as needed	As above via PO, NG or IV
Severe dehydration, adolescents and adults	20ml/kg IV crystalloid titrated to pulse rate	As above via PO, NG or IV

Infants with diarrhea should continue to be breastfed

PHARMACOLOGICAL MEASURES- ACUTE DIARRHEA

Empiric antibiotic therapy is not recommended for acute diarrhea (unless travellers' diarrhea): self-limiting, antibiotic resistance & side effects

Bismuth
subsalicylate
[Pepto
Bismol®]

- Frequently used in acute diarrhea; no clear evidence of its efficacy
- Exhibits antisecretory, antimicrobial and some anti-inflammatory activity
- Salicylate component can cause gastric and duodenal damage; black stools (can be mistaken for melena)
- Dose: 30 mL q 30 mins up to a maximum of 8 doses/day
- **CAUTION:** If someone has a true allergy to Aspirin®, they could develop an allergic response to Pepto-Bismol®. Risk of Reye's syndrome in children and adolescents < 18 years of age with or recovering from chicken pox or viral infection with flu-like symptoms.
- **Avoid in children < 12 years of age and in pregnancy and breastfeeding; caution in renal impairment; avoid in patients taking anticoagulants, methotrexate, and salicylates**

PHARMACOLOGICAL MEASURES- ACUTE DIARRHEA

Empiric antibiotic therapy is not recommended for acute diarrhea (unless travellers' diarrhea): self-limiting, antibiotic resistance & side effects

Opioids
(codeine,
diphenoxylate,
loperamide)

- Codeine: doses necessary may not be safely attainable with acetaminophen-containing non-prescription products and may result in acetaminophen toxicity
- Diphenoxylate: less effective than loperamide, may cause anticholinergic side effects
- **Imodium® (loperamide):**
 - ☐ **Avoid** in children < 12 years of age; **contraindicated** in children < 2 years
 - ☐ Not recommended for use in patients with bloody diarrhea or fever (i.e. infectious)
 - ☐ **May be used** in pregnancy and breastfeeding
 - ☐ **First choice** in patients with uncomplicated chemotherapy-related diarrhea
 - ☐ **> 12 years old: 4 mg STAT, then 2 mg PO after each loose BM. Maximum 16 mg/day**
- D/C if symptoms of diarrhea persist > 48 hours
- ADRs: cramps, dry mouth and drowsiness, dysrhythmias (when used at higher than recommended doses), opiate-like CNS effects observed in children < 3 years
- **Use with caution** in patients with a history of opioid use disorder

PHARMACOLOGICAL MEASURES- CHRONIC DIARRHEA

Opioids (codeine, diphenoxylate, loperamide)	• Effective for symptomatic use • Tolerance to adverse effects (e.g. sedation) over time with chronic use • Loperamide well tolerated, available OTC, lowest abuse potential • Reports of abuse at supratherapeutic doses to achieve CNS opioid effects • Codeine has most CNS side effects compared to loperamide or diphenoxylate
Psyllium (Metamucil®)	• Limited role • Avoid combinations with laxatives & ensure adequate hydration to avoid esophageal or intestinal obstruction • Dose: 1 teaspoon Q12H PO
Cholestyramine (Olestyr®)	• Useful for bile-acid induced diarrhea secondary to malabsorption of bile acids • Example: Crohn's Disease, Irritable Bowel Syndrome • Dose: 4g Q12H PO • Important: Other medications to be administered 1 hour before or 4-6 hours after. Ensure adequate hydration. • May cause nausea and fat-soluble vitamin deficiencies
Clonidine	• Treatment for diarrhea associated with opioid withdrawal • Doses required may cause sedation and dry mouth. Additionally, can precipitate hypotension when used concomitantly with hypotensive drugs.
Octreotide	• Treatment for diarrhea caused by neuroendocrine tumours

SPECIAL POPULATIONS- PREGNANCY AND BREASTFEEDING

Pregnancy

- Bulk-forming agents (e.g. psyllium) are first-line
- Loperamide has limited human pregnancy data: may be used as second-line option when risk of dehydration outweighs risk of drug used
- **DO NOT USE:** diphenoxylate/atropine* (Lomotil®), bismuth subsalicylate

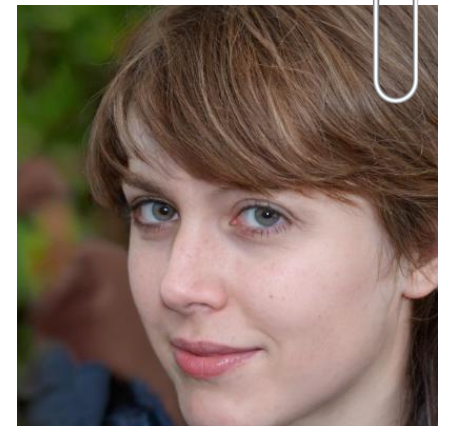
Breastfeeding

- Breastfeeding should be continued in infants with diarrhea
- Loperamide or bulk-forming agents may be used
- **DO NOT USE:** diphenoxylate/atropine* (Lomotil®), bismuth subsalicylate (salicylates transferred to breast milk)

*Atropine is added to diphenoxylate to reduce abuse potential

CASE 1

DR is a 23-year-old female who has been experiencing diarrhea over the past 2 days. She has been having approximately 4 bowel movements per day, whereas she normally only has one. She has not travelled outside the country in the past month and does not have a fever, nausea/vomiting, or bloody stools. She does not have any drug allergies. Her medical conditions include menstrual cramps and asthma. She is currently taking Marvelon-28® (desogestrel/ethinyl estradiol 0.15/0.03 mg), salbutamol 100 mcg (1 puff inhaled q 4-6 hours) PRN, Advair® (fluticasone/salmeterol) 25/250 mcg 1 puff inhaled BID, naproxen 220 mg PO BID PRN for occasional headaches. Additionally, DR has been using magnesium oxide for restless legs syndrome (RLS) for the last 3 days. She smokes socially, typically 1-2 cigarettes over the weekend.



CASE 1 - QUESTIONS

1. Which type of diarrhea is DR most likely experiencing?
 - a) Infectious
 - b) Inflammatory
 - c) Medication-induced
 - d) Malabsorption

2. Which of the following medications is most likely contributing to her diarrhea?
 - a) Desogestrel/ethinyl estradiol 0.15/0.03 mg (Marvelon® -28)
 - b) Magnesium oxide
 - c) Salbutamol
 - d) Fluticasone/salmeterol 25/250 mcg (Advair®)

CASE 1 - QUESTIONS

3. Which of the following would NOT be an appropriate non-pharmacological recommendation for DR?
 - a) Advise DR to avoid foods containing sorbitol, mannitol or xylitol
 - b) Advise DR to decrease her coffee consumption
 - c) Advise DR to increase consumption of bananas, rice, unsweetened applesauce and toast until diarrhea resolves
 - d) Advise DR to increase consumption of lactose-containing dairy products

4. Which of the following oral rehydration therapies would you recommend to DR?
 - a) A balanced sodium-to-magnesium ratio
 - b) A balanced calcium-to-sodium ratio
 - c) A balanced sodium-to-glucose ratio
 - d) A balanced calcium-to-glucose ratio

INFECTIOUS DIARRRHEA

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INFECTIOUS DIARRHEA - OVERVIEW

- Diarrhea caused by microbiologic agents: bacteria, viruses, parasites or toxins

Clinical finding	Salmonella species	Shigella species	Campylobacter species	C. difficile	Entamoeba histolytica
Fever	√	√	√	+/-	√
Abdominal pain	√	√	√	√	
Vomiting/ Nausea	√	√	√	+/-	
Bloody stool	√	√	√	ABSENT!	
Tenesmus		√			
Fecal WBC		√	√	√	√
Fecal lactoferrin	√	√		√	

WHAT ARE ENTERIC INFECTIONS?

Enteric infections

Non-bloody (e.g. enterotoxin-producing, cholera) diarrhea

- Characterized by watery diarrhea with no increase in fecal leukocytes
- Small bowel is affected
- In neonates (6-18 months), it is often caused by rotavirus
- Healthy adults in hygienic environment often caused by food poisoning, parasites, viruses
- Adults in unhygienic environments: *Vibrio cholerae*, enterotoxigenic *E. coli*
- Travellers' diarrhea: enterotoxigenic *E. coli*, *Salmonella* spp.

Bloody (e.g. enterohemorrhagic, dysentery) diarrhea

- Characterized by mucus or blood in the feces with an increase in fecal mononuclear leukocytes.
- Travellers' diarrhea often caused by *Salmonella* spp., *Shigella* spp., *Campylobacter* spp., Enterohemorrhagic *E. coli* (EHEC/STEC)
- Non-travel-related diarrhea often caused by *Yersinia* spp., *Salmonella* spp., *Shigella* spp.

GENERAL GUIDELINES TO TREATING INFECTIOUS DIARRHEA

- Intent to treat with antibiotics must be carefully weighed against increased risk of antimicrobial resistance
- Treat only after fecal sample obtained*
 - Exception: Travellers' diarrhea (toxigenic *E. coli*)
 - Pre-emptive treatment recommended at start of symptoms
 - Reduces illness duration and severity
- Empirical therapy should be considered if symptoms of febrile diarrhea or severe symptoms (e.g. hypotension)
 - Fluoroquinolones or macrolide (azithromycin) for adults
 - Trimethoprim/Sulfamethoxazole (TMP-SMZ) or macrolide for children
 - Metronidazole for parasitic infection (*Giardia*)

GENERAL GUIDELINES TO TREATING INFECTIOUS DIARRHEA

Use of anti-diarrheal agents (e.g. loperamide)

- Only for use in adults with mild, non-bloody diarrhea (travel or non-travel related)
- Should be avoided in patients with bloody and/or moderate to severe diarrhea
- Use lowest effective dose, monitor for post-diarrheal constipation
- Do not use in *C. difficile* disease or those with toxin-mediated disease

TRAVELLERS' DIARRHEA

Severity	Characteristics
Mild	• Tolerable • Non-distressing • Does not interfere with planned activities
Moderate	• Distressing • Interferes with planned activities
Severe	• Incapacitating, completely stopping all planned activities or • Dysentery diarrhea (e.g. bloody stool)
Persistent	• Diarrhea lasting ≥ 2 weeks

PHARMACOLOGICAL MEASURES- TRAVELLERS' DIARRHEA

Self-treatment	• Mild diarrhea → tolerable and does not interfere with daily activities • ORT (oral rehydration therapy) • Bismuth Subsalicylate • Loperamide = not recommended to be used if bloody diarrhea or fever (> 38.5°C)
Antibiotics in Travellers' Diarrhea	• Not recommended for most adults with non-severe, watery diarrhea. • Used for moderate-severe travellers' diarrhea • Diarrhea is interfering with activities; high fever; bloody or mucoid stools • Ciprofloxacin, Levofloxacin, Norfloxacin → Latin America & Africa • Azithromycin → Asia & Mexico [Preferred in children and pregnant/breastfeeding women] • If antibiotic therapy used in last 3 months and no travel history, assess for C. diff
Vaccines	• DUKORAL®: Prevention of TD caused by enterotoxigenic E.coli diarrhea. • Not for all travellers • May be used in those with high risk of complications or in those with serious inconveniences from TD (athletes, politicians)

PATHOGENESIS FOR *C. DIFFICILE* INFECTION (CDI)

Susceptible host

- One or more risk factors



Loss of protective flora

- Antibiotic exposure and/or decreased stomach acidity



Exposure to *C. difficile* spores

- Bedside items, caregiver's hands



Ingestion of spores

- Transmission via fecal-oral route



Asymptomatic colonization

- *C. difficile* colonization

C. DIFFICILE

Detection

- Stool culture (expensive) – not frequently used
- Immunoassay testing for toxin A & B is faster
- Presence of pseudomembranous colitis.

Risk Factors

- Antibiotic usage (most common): Clindamycin, 3rd and 4th generation cephalosporins, FQ, broad-spectrum penicillins
- Hospitalization
- Severe illness
- Chemotherapy
- Older adults
- Immunocompromised
- PPI/H2RA usage

Symptoms

- Frequent loose/watery diarrhea (≥ 3 loose stools/24 hours)
- Blood might be present (rare)
- Abdominal pain/tenderness
- Nausea/vomiting
- Fever (> 38.5); foul smell

CDI – MANAGEMENT

- Discontinue other antibiotics if possible
- **Important counselling point:** Highest risk of acquiring *C. diff.* is during the first month after cessation of an antibiotic.
- Discontinue anti-diarrheals
- Discontinue proton pump inhibitors/ H₂ antagonists if not indicated (may help reduce recurrence, unlikely to impact acute treatment)
- Start antibiotics targeted towards *C. difficile*

Note, antibiotics are NOT recommended for preventing CDI

CDI – TREATMENT

STEP #1: D/C antibiotic agent the patient is on

STEP#2: Manage fluid loss

STEP 3: Treatment/Eradication

Non-severe disease (White blood cell count $\leq 15,000$ cells/ μL and serum creatinine $< 133 \mu\text{mol/L}$)

- Can be managed in outpatient settings
 - **Preferred agent:** Fidaxomicin 200mg PO BID for 10 days
 - **Alternative Agents:** Vancomycin 125 mg QID PO x 10 days
 - If vancomycin or fidaxomicin unavailable: Metronidazole 500 mg po TID for 10 days

Severe disease (White blood cell count $> 15,000$ cells/ μL and/or serum creatinine $\geq 133 \mu\text{mol/L}$)

- Very ill & often managed in hospital settings
 - **Preferred agent:** Fidaxomicin 200mg PO BID for 10 days
 - **Alternative Agents:** Vancomycin 125 mg QID PO x 10 days

Fulminant disease (Meets criteria for severe disease **PLUS hypotension OR shock OR toxic megacolon OR ileus**)

- **Preferred agent:** Vancomycin 500 mg PO QID. **Consider adding** Metronidazole 500 mg Q8H IV **if ileus is present**
- **Alternative Agents:** If ileus is present, may add Vancomycin 500 mg in 100 mL saline Q6H PR as retention enema

CDI – RECURRENCE

FIRST RECURRENCE (10-20% of cases), within 8 weeks of completing treatment

- **Preferred agent:** Fidaxomicin 200 mg BID x 10 days
- **Alternative Agents:** Vancomycin in a tapered and pulsed regimen
- If metronidazole was used in the 1st episode, consider the standard vancomycin course of therapy

What is Fecal Microbiota Transplantation (FMT)?

- It is the process where stool from a relatively healthy individual is transferred to a sick patient to help prevent recurrent CDI
- Colonoscopy or rectal enema suspension
- Risk: infection transmission

PREVENTION OF INFECTION

Hand hygiene

- Wash hands with soap and water instead of disinfectant since spores are not killed by disinfectant
- Use of gloves and gowns for health care workers and visitors

Early identification

Pre-emptive isolation

- Contact precautions for duration of diarrhea

Environmental disinfection

- Replace equipment with disposables where possible
- Chlorine containing cleaning agents

Single patient rooms → Isolate patient

Antimicrobial use restriction

- Implementation of antimicrobial stewardship program

Probiotics are NOT routinely recommended

TAKEAWAY

- Diarrhea may be **acute (< 14 days)**, **persistent (14 days-4 weeks)** or **chronic (> 4 weeks)**
 - Although typically mild and self-limited, diarrhea can lead to **dehydration, which can be life-threatening**
 - **Oral Rehydration Therapy (ORT)** is the gold standard for outpatient management of dehydration
 - Pepto Bismol® (bismuth subsalicylate) should be **avoided in those with an ASA allergy**
 - Loperamide should be **avoided in children < 12 years of age** and in those with **bloody diarrhea/fever**. Use with caution in patients with a **history of opioid use disorder**
- Travellers' Diarrhea:
 - Mild → can be managed with **ORT, bismuth subsalicylate, loperamide**
 - Moderate-severe → may warrant antibiotic therapy
- *C. difficile*-associated diarrhea:
 - **Antibiotic use and hospitalization** are common risk factors for the development of *C. difficile* diarrhea
 - **Non-fulminant CDI** (initial episode and initial recurrence) – **fidaxomicin** is the preferred agent. Vancomycin can be used as an alternative
 - **Fulminant CDI** – **vancomycin** is the preferred agent.

CASE 2

ID is a 49-year-old female who presents to your local community hospital. She has been experiencing 5-6 watery, loose BMs/day for the past 2 days. At triage, her temperature was 38.9°C. ID recently had a dental surgery and was prescribed clindamycin 300 mg PO QID for infection prophylaxis. She has been taking the antibiotic for the past 4 days. She has had no recent travel. ID's medical history is significant for gastroesophageal reflux disease (GERD)(3-year diagnosis), for which she takes esomeprazole 20 mg PO daily and Tums® PRN. She also takes 1000 units of Vitamin D3 daily and ibuprofen 400 mg PRN for occasional headaches (1-2 times per month). ID does not smoke or take recreational drugs, but drinks alcohol occasionally. She has a documented allergy to penicillin, having experienced a rash as a child.

You have access to Clinical Viewer, and note the following findings:

BP = 124/70 mmHg

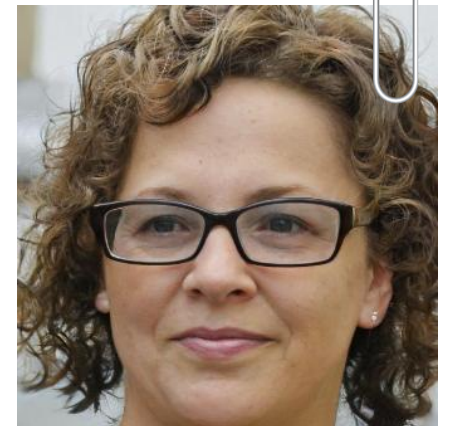
HR = 110 bpm

WBC = $17.6 \times 10^9/\text{L}$ ($3.5\text{-}10.5 \times 10^9/\text{L}$)

Lactate = 2.1 mmol/L (0.5-2.5 mmol/L)

Albumin = 36.1 g/L (34-50 g/L)

SCr = 115 $\mu\text{mol/L}$ (baseline = 65 $\mu\text{mol/L}$).



CASE 2 - QUESTIONS

1. Which investigation(s) would be most appropriate at this time?
 - a) Abdominal exam
 - b) Stool testing for *C. difficile*
 - c) Stool culture
 - d) Endoscopy
2. Which of the following factors increases the patient's CDI risk the most?
 - a) Use of esomeprazole
 - b) Use of clindamycin
 - c) Use of ibuprofen
 - d) Alcohol consumption
3. How would you classify the severity of ID's diarrhea?
 - a) Mild-moderate
 - b) Severe
 - c) Fulminant
 - d) Recurrent
4. Which of the following regimens would be most appropriate for ID?
 - a) Metronidazole 500 mg PO TID x 14 days
 - b) Vancomycin 125 mg PO QID x 10 days
 - c) Vancomycin 250 mg PO QID x 10 days
 - d) Fidaxomicin 200 mg PO once daily x 10 days

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CHANGE LOG

June 2025

- Slide 12: Updated caution info for bismuth to include information on risk of Reye's syndrome

August 2025

- Slide 3: Legend updated
- Combined and consolidated information on slide 20
- Rearranged slides on Traveller's Diarrhea for better flow
- Slides 29 and 30: updated to reflect guidelines
- Spelling changes throughout

September 2025

- Slide 34: Updated option D in question 4 to make option B the only correct option

October 2025

- Slide 13: Updated serum creatinine threshold to 133 umol/L to match IDSA guidelines

November 2025

- Minor grammar and wording changes throughout
- Slide 3: Updated legend
- Slide 32: Updated takeaway
- Slide 34: Question 2, updated wording and answer choices

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- Slide 12: added viral infection
- Slide 13: Minor update that diphenoxylate can cause *anticholinergic* side effects
- Slide 30: Changed oral capsule to rectal enema suspension (Rebyota® monograph, Ref 21)
- Slide 35: Updated references